



TO: _____ North Carolina Community Action Agencies

FROM: _____ Valerie Powell
_____ CSBG Program Manager

DATE: _____ April 7, 2020

RE: _____ Relief NC Application for Funding

The NC Department of Health and Human Services, Office of Economic Opportunity (OEO) was awarded \$8 million from the US Office of Community Services (OCS) to administer disaster supplemental funds to designated NC Community Action Agencies (CAAs). Funding supports a range of locally identified services and strategies focused on residents with low incomes and disaster-related needs.

OEO's disaster supplemental funds program, called Relief NC, specifically targets individuals and families that have unmet needs as a result of Hurricane Florence that occurred in 2018. Eligible funding domains are employment, education, cognitive development, infrastructure, housing, health and social/behavior development, transportation, home repair and emergency assistance.

CAAs that serve counties declared a Presidential disaster due to Hurricane Florence are encouraged to apply for Relief NC funding. If you elect not to apply for the Relief NC funding, please contact me as soon as possible as your agency's disaster allocation can be redistributed.

Applications for Relief NC must be e-mailed to Doris McLamb at Doris.McLamb@dhhs.nc.gov and carbon copied to Valerie.Powell@dhhs.nc.gov by Monday, April 27, 2020. If you have questions regarding the Relief NC, contact me at 919-820-4636.

Attachments: _____ Application Instructions
_____ Budget Form

~~Introduction~~ **ENGINEERS AND GEOSCIENTISTS BC**
2004 BIOMEDICAL / BIOCHEMICAL SYLLABUS
For Self-Evaluation

Name: _____

User _____

ID: _____

For directions, refer to the Instructions for Completing Syllabus and Course Descriptions.

~~Regulations outlined in CSBG 45 CFR 96 are applicable to the Relief NC Program. Additional guidance from OCS will be forthcoming to specifically address disaster related activities.~~

~~For the purpose of the Relief NC Program, OEO, in consultation with the North Carolina Community Action Association, established additional policies regarding the administration of the Relief NC Program. Eligible CAAs should review CSBG 45 CFR 96 and the information outlined below for policy guidance.~~

- ~~1. Relief NC funding shall not be available for costs that are reimbursed by the Federal Emergency Management Agency (FEMA), under a contract for insurance or by self insurance. Funding is for non-duplication of FEMA assistance and non-duplication for life, health, automobile, homeowners, renters and property insurance.~~
- ~~2. Participants eligible for Relief NC funding must self-certify and attest that proposed services and activities were not covered by FEMA or insurance. In addition, participants must attest that falsification or misrepresentation may result in the rejection of application and may be subject to prosecution under applicable State and Federal statutes.~~
- ~~3. Eligible domains for funding are:~~

~~● Employment Projects~~

~~Employment projects are designed to place low-income clients in permanent jobs (minimum of 30 hours per week) paying at least the minimum wage. A permanent job is anticipated to be year-round with no predetermined ending date.~~

~~● Education Projects~~

~~Education projects are designed to provide activities that improve the education or skills level of low-income individuals through the completion of a formal, certified education or skills training program. Education programs should improve the employability or income-earning potential of participants.~~

- ~~Cognitive Development Projects~~

~~Cognitive development projects are intended to help participants obtain outcomes in the Education and Cognitive Development Domains. These services include: child/young adult education programs; school supplies; extracurricular programs; adult education programs; post-secondary education supports; financial aid assistance and home visits.~~

- ~~Infrastructure Projects~~

~~Infrastructure projects represent the basic physical and organizational structures and facilities needed to coordinate and enhance multiple programs and resources that address poverty conditions in local communities.~~

- ~~Housing Projects~~

~~Housing projects are designed to move low-income families from a sub-standard housing condition to a standard one. Funds are prohibited for the purchase, construction or permanent improvement, other than low-cost residential weatherization or other energy-related home repair, of any building or other facility.~~

- ~~Health & Social/Behavior Development Projects~~

~~Health & Social/Behavior projects improve a participant's physical, mental, or behavioral health; increase a participating household members' interaction skill; demonstrate a participant's ability to live independently and decrease a participating household member's recidivism rate.~~

- ~~Transportation Projects~~

~~Transportation projects consists of gas vouchers, bus passes and car repairs for employment or education services.~~

- ~~Home Repair Projects~~

~~Home repair projects provide property improvements that protect or improve the basic livability or utility of property. For the purposes of the Relief NC Program, home repairs provided by the program cannot improve the tax value of the home.~~

~~Individual and combined repairs exceeding \$1,000 per household requires three quotes from a contractor licensed by the State of North Carolina. The selected licensed contractor must enter into a contractual agreement between the CAA and the licensed contractor.~~

~~Individual or combined home repair activities that do not exceed \$1,000 per household may be provided by a licensed contractor or carpenter approved by the CAA.~~

- ~~Emergency Assistance Projects~~

~~Emergency assistance projects provide direct assistance to low-income families who are experiencing emergency situations in the areas of food, shelter, fuel, utility cut-off and medicine.~~

Community Services Block Grant Program Relief NC Application

Please save as a PDF document and upload via your applicant portal.

Agency Information

<u>Exam</u>	<u>DUNS Number</u> :	<u>Exa m Na me</u>	<u>Applicant's Self-Evaluation - Course Equivalent Code</u>	<u>Page Number Reference</u>	<u>For O</u>
<u>Basic Studies (7 Required)</u> Administrative Office Address:					
<u>04- BS -1</u>	<u>Mat hem atics</u>				
<u>04- BS -5</u>	<u>Adv anc ed Mat hem atics</u>				

04- BS -7	Mechanics of Fluids		
04- BS -10	Thermodynamics		
04- BS -11	Properties of Materials		
04- BS -12	Organic Chemistry		
04- BS -13	Biology		
Basic Studies (2 required)			
04- BS -2	Probability & Statistics		

04-BS-3	Statics and Dynamics				
04-BS-4	Mailing Address (include the 4-digit zip code extension):	Electric Circuits and Power			Full Credit Comments
04-BS-6	Telephone Number:	Mechanics of Materials			Full Credit Comments
04-BS-14	Fax Number:	Geology			Full Credit Comments
04-BS-15	Engineering Graphics & Design Process Disaster Fund Allocation \$	Relief NC			Full Credit Comments
Proposed Funding:					
Group A (6 required)		Application Period:	Beginning: TBD		Ending: TBD

04-Bio-A1 Board Chairperson:		Bio mat erial s and Bio c omp atibil ity			<u>Full Cre</u> <u>Comments</u>
04-Bio-A2 Board Chairperson's Address: (where communications should be sent)		Proc ess Dyn amic s and Cont rol			<u>Full Cre</u> <u>Comments</u>
04- Bio -A3	Cellular and Molecular Biology and Biochemistry Board Chairperson's Term of Office (enter beginning and end dates):				<u>Full Cre</u> <u>Comments</u>
04-Bio-A4 Executive Director:		Bio mec hani cs			<u>Full Cre</u> <u>Comments</u>
04-Bio-A5 Executive Director Email Address:		Enz yme and Micr obial Kine tics			<u>Full Cre</u> <u>Comments</u>
04-Bio-A6 Agency Fiscal Officer:		Anat omy and Phy siolo gy			<u>Full Cre</u> <u>Credit</u> <u>Comments</u>
04-Bio-A7 Fiscal Officer Email Address:		Flui d Mec hani cs			<u>Full Cre</u> <u>Comments</u>

04-Bio-A8CSBG Program Director:		Biop hysi cal Mea sure men ts			Full Cre Comments
04-Bio-A9CSBG Program Director Email Address:		Bior eact or Desi gn			Full Cre Comments
Group B (3 Required) Counties Served with Relief NC funds:					
04-Bio-B1 Agency Operational Fiscal Year:		Bioc hem ical Sep arati ons			Full Cre Comments
04-Bio-B2	Pros thes es and Orth oses				
04-Bio-B3	Biotr ans port Phe nom ena				
04-Bio-B4	Digit al Ima ge				

	<u>Proc</u> <u>essi</u> <u>ng</u>		
<u>04-</u> <u>Bio</u> <u>-B5</u>	<u>Cell</u> <u>and</u> <u>Tiss</u> <u>ue</u> <u>Engi</u> <u>neer</u> <u>ing</u>		
<u>04-</u> <u>Bio</u> <u>-B6</u>	<u>Bioi</u> <u>nstr</u> <u>ume</u> <u>ntati</u> <u>on</u>		
<u>04-</u> <u>Bio</u> <u>-B7</u>	<u>Rob</u> <u>otics</u> <u>and</u> <u>Man</u> <u>ufac</u> <u>turin</u> <u>g</u> <u>Auto</u> <u>mati</u> <u>on</u>		
<u>04-</u> <u>Bio</u> <u>-B8</u>	<u>Reh</u> <u>abilit</u> <u>atio</u> <u>n</u> <u>Engi</u> <u>neer</u> <u>ing</u>		
<u>04-</u> <u>Bio</u> <u>-B9</u>	<u>Artifi</u> <u>cial</u> <u>Intell</u> <u>igen</u> <u>ce</u> <u>and</u>		

	<u>Expert Systems</u>		
<u>04-Bio - B10</u>	<u>Analytical Biochemistry</u>		
<u>04-Bio - B11</u>	<u>Ergonomics</u>		
<u>04-Bio - B12</u>	<u>Applied Optics/Photonics</u>		
Complementary Studies (All Required)			
<u>11-CS -1</u>	<u>Engineering Economics</u>		
<u>11-CS -2</u>	<u>Engineering in Society = Health and</u>		

	<u>Safe</u> <u>ty</u>		
<u>11-</u> <u>CS</u> <u>-3</u>	<u>Sust</u> <u>aina</u> <u>bility</u> <u>Engi</u> <u>neer</u> <u>ing</u> <u>and</u> <u>the</u> <u>Envi</u> <u>ron</u> <u>men</u> <u>t</u>		
<u>11-</u> <u>CS</u> <u>-4</u>	<u>Engi</u> <u>neer</u> <u>ing</u> <u>Man</u> <u>age</u> <u>men</u> <u>t</u>		

North Carolina Department of Health and Human Services
Office of Economic Opportunity—
2420 Mail Service Center / Raleigh, North Carolina 27699-2420

**Relief NC Program
Application for Funding
Certification and Assurances**

Public Hearing on the Initial Plan

We herein certify that a public hearing as required by 10A NCAC 97B .0402 Citizen Participation in the Application Process occurred on _____ for the planning process for the Relief NC Program and the agency has maintained documentation to confirm the process of the public hearing.

For multi-county providers, indicate the date and the county the hearing was held. _____

Date	County	Date	County

Board of Directors Approval of the Application

I hereby certify that the information contained in the attached application is true and the Board of Directors has reviewed and approved this application for the Relief NC Program.

Date of Board Approval: _____

Board Chairperson: _____

(Signature) (Date)

Finance Committee Chairperson: _____

(Signature) (Date)

Relief NC Program
Documentation of Submission to County Commissioners

Community Action Agencies receiving Relief NC funding must provide a completed copy of the Application for Funding to local boards of commission and submit documentation of delivery (return mail receipt, signature of county staff noting receipt, electronic mail acknowledging receipt) to the Office of Economic Opportunity.

Agency Name: _____

County: _____

Date of Application Submission: _____

_____ The agency submitted a complete application for funding to local Board(s) of Commission.

_____ CAA attached documentation confirming local Board(s) of Commission received Relief NC Application for Funding.

Section 1: Community Needs Assessment

A community needs assessment is a systematic process for determining and addressing needs, or "gaps" between current conditions and desired conditions or "wants." From the list below, identify how your agency will determine the unmet needs of your county.

(Check one or more of the following methods)

- ☐ ~~Surveys of the community(s) – door to door, telephone, etc.~~
- ☐ ~~Review of Records – agency intake forms, program participant records, etc. (may be used with at least one other type of needs assessment; will not meet compliance on its own)~~
- ☐ ~~Review of demographical information – U.S. Census, welfare statistics, unemployment statistics, etc.~~
- ☐ ~~Discussions/information/testimony provided by individuals and community members – social service professionals, agency staff, program participants, etc.~~
- ☐ ~~Public meetings to solicit input on community needs~~
- ☐ ~~Other~~

(Describe) _____

Please

BIOMEDICAL/BIOCHEMICAL ENGINEERING EXAMINATIONS

GROUP A - COMPULSORY EXAMINATIONS (SIX REQUIRED)

04-Bio-A1 - Biomaterials and Biocompatibility

Structure and properties of amorphous solids. Physical and chemical bases for properties exhibited by materials. Polymeric biomaterials. Metallic biomaterials. Ceramic biomaterials. Composite materials. Material properties including mechanical, electrical, magnetic and thermal behaviour. Applications of biomaterials in tissue and organ systems. Relationship between physical and chemical structure of materials and biological system response. Selection, fabrication and modification of materials for specific biomedical applications. Biomaterials processing. Biomaterials degradation. Implant requirements. Host-implants reactions including wound healing response and inflammatory response. Physiological and biomechanical basis for soft-tissue implants. Design of modified biomaterials. Bulk and surface characterization of materials. Regulatory and ethical concerns dealing with the implementation and commercialisation of biomaterials and medical devices.

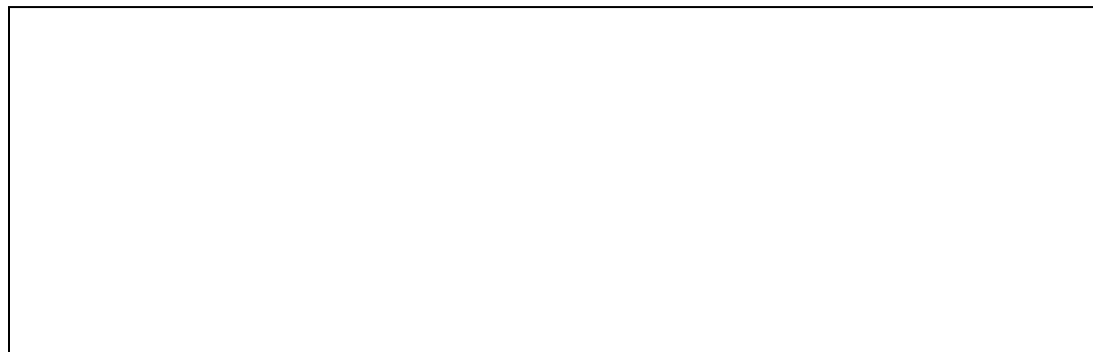
Suggested Text:

Ratner, Buddy DS., Allan S. Hoffman, Frederick J. Schoen and Jack E. Lemons editors, Biomaterials Science, An ~~submit documents to demonstrate how residents participated in the community needs assessment process in your county(s).~~

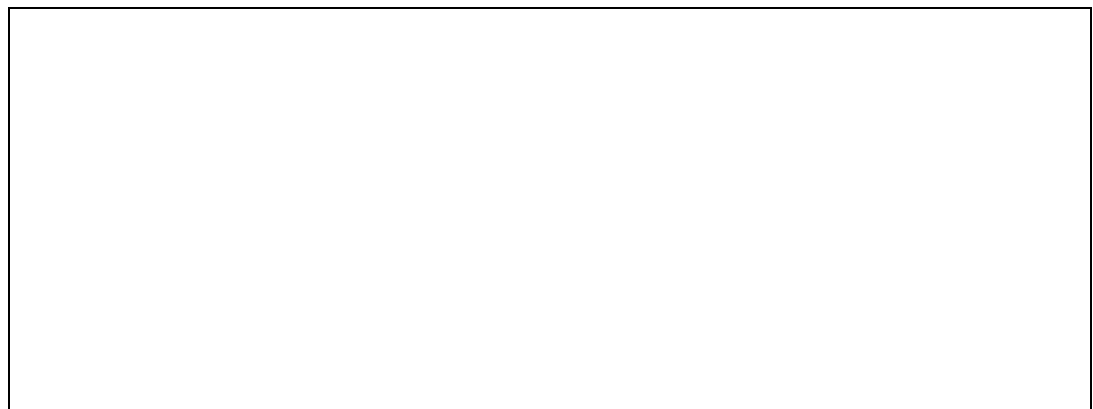
From the data collected using the method(s) identified in Section 1, describe the unmet employment needs identified.

From the data collected using the method(s) identified in Section 1., describe the unmet education needs identified.

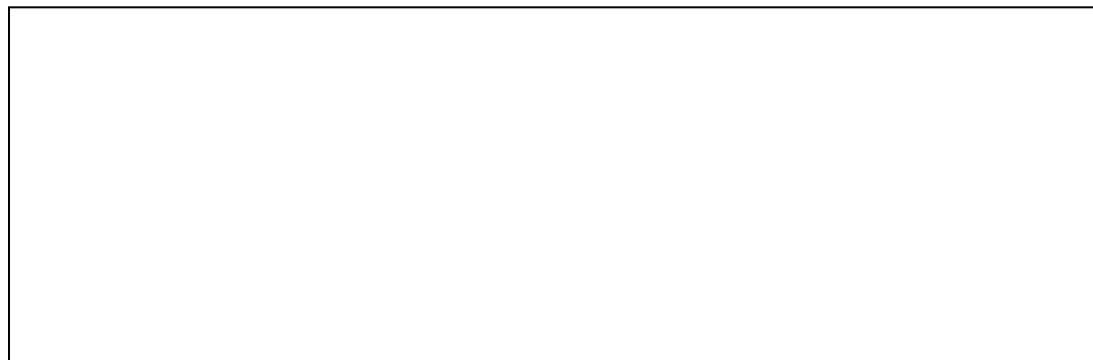
From the data collected using the method(s) identified in Section 1., describe the unmet cognitive development needs identified.



~~From the data collected using the method(s) identified in Section 1., describe unmet needs identified.~~



~~From the data collected using the method(s) identified in Section 1., describe the unmet housing needs identified.~~



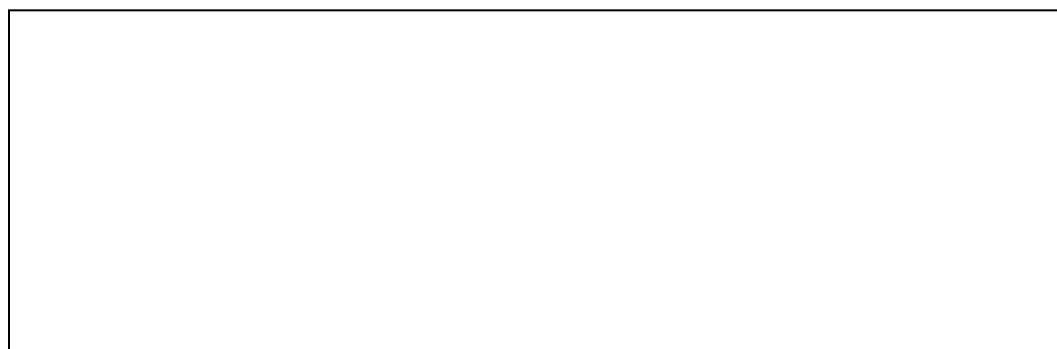
From the data collected using the method(s) identified in Section 1., describe the unmet health and social/behavioral development needs identified.



From the data collected using the method(s) identified in Section 1., describe the unmet transportation needs identified.



~~From the data collected using the method(s) identified in Section 1., describe the unmet home repairs needs identified.~~



~~From the data collected using the method(s) identified in Section 1., describe the unmet and emergency assistance needs identified.~~

Section 2: Prioritization of Needs

Rate the unmet needs using a scale of 1 – 9 whereas 1 represents the greatest need and 9 represents a lesser priority (list rating beside each domain name).

Employment—	Education
Cognitive Development	Infrastructure
Housing	Health & Social/Behavior Development
Transportation——	Home Repair
Emergency Assistance	

Section 3: Outreach to Local Communities

Describe how your agency will conduct outreach opportunities to residents with unmet needs as a result of Hurricane Florence.

Will your agency provide enhanced outreach to special populations (key an “x” beside appropriate response)?

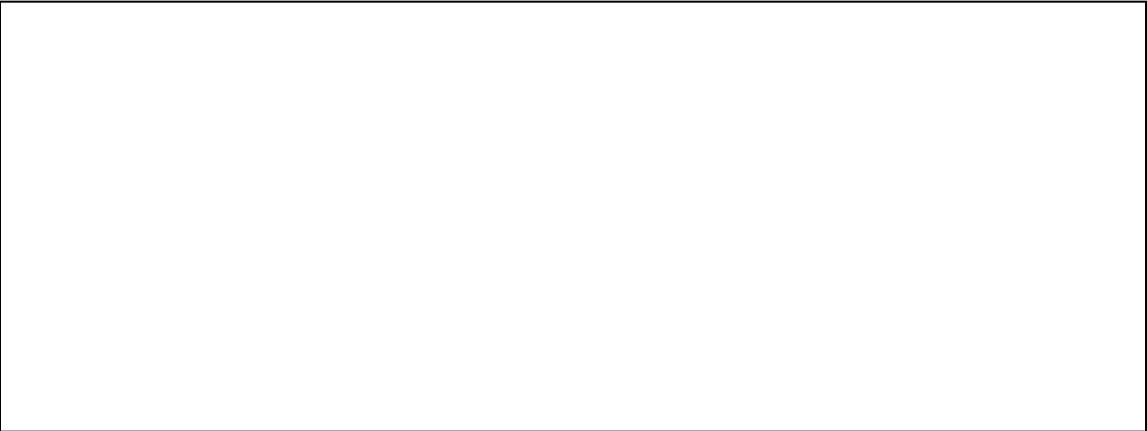
Yes _____ No

If yes, what are the special populations to be served i.e. senior citizens, undocumented residents, homeless, disabled, etc.).

Describe how enhanced outreach opportunities will be provided to special populations.

Section 4: Prioritize Clients Served

Describe the process how your agency will prioritize clients to be served. The process should include consideration of clients with the greatest barriers.



Section 5: Project Identification

1. Project Name:				
2. Objective Statement:				
3. Relief NC Funds Requested for this Project:				
4. Project Period:	TBD	To	TBD	
5. Total Number Expected to Be Served:				
-Section 6: One-Year CSBG Program Objective and Activities				
Identified Problem	Service or Activity	Outcome Expected	NPIs (List all NPIs applicable to activity)	Position Title(s)

Use the table below to enter your agency's targeted outcome results. The performance measures will be included in the agency's Relief NC contract.

CSBG grantees operating Relief NC projects are required to enter program targets in Table 1. Grantees have the flexibility to enter specific program targets based upon the community needs assessment.

Table 1 Outcome Measures for Project 1 (enter project name)	
Measure	Expected to Achieve the Outcome in Reporting Period (Target)
The number of participant families served.	
The number of low-income participant families rising above the poverty level.	
The number of participant families obtaining employment.	
The number of participant families who are employed and obtain better employment.	
The number of jobs with medical benefits obtained.	
The number of participant families completing education/training programs.	
The number of participant families securing standard housing.	
The number of participant families provided emergency	

assistance.	
The number of participant families provided employment supports.	
The number of participant families provided educational supports.	

CONTRACT BUDGET NARRATIVE STATE OF NORTH CAROLINA DIVISION OF SOCIAL SERVICES OFFICE OF ECONOMIC OPPORTUNITY Form 6844N	
Name of Agency:	
Section A – Salaries and Wages	
Section B – Fringe Benefits	
Section C – Equipment Purchases	
Section D – Communication	
Section E – Space costs	
Section F – Travel/Employee Development	
Section G – Supplies and Materials	
Section H – Contractual Services	
Section I – Client Services	
Section J – N/A	

Agency Name

Section K — Other
Section L — Indirect Costs

~~Checklist to Submit a Complete Relief NC Application~~

Please place a check mark in the appropriate box to show that you have included the completed documents with your application

Item	Included (✓)
Signed Application Certification (blue ink only)	
Board of Directors Approval Statement	
Documentation of Submission to Local Board(s) of Commission	
Community Needs Assessment Documents:	
• Documentation of Residents Participation In Community Assessment Process	
• Policy Describing How Clients Will Be Prioritized To Receive Services	
Form 212 — One Year Work Program	
Budget and Narrative	
Job Descriptions For All New CSBG Employees (for disaster project)	
Documentation of Public Hearings for Initial Planning Process:	
Copy of Public Notice(s) From Newspaper(s)	
Agenda of Public Meeting(s)	
Copy of Attendance Sheet(s)	

Minutes of Public Meeting(s)	
Documentation for Notice of Intent to Apply	
Copy of Advertisement(s)	
Commissioners' Comments or Minutes (if applicable)	

Introduction to Materials in Medicine. Academic Press, NY, 1996.

04-Bio-A2 - Process Dynamics and Control

Linear models of physical systems and processes, the concept of the transfer function. The transient response of linear systems to step, ramp and sinusoidal inputs. Bode plots and the frequency response analysis of systems. On-off, proportional, integral, derivative and combined control actions. Stability analysis of closed-loop systems using the root locus method and the Nyquist criterion. Feedback and feedforward control. The state-space analysis of control systems. Modeling of nonlinear systems using the phase-plane and describing functions methods, stability of control systems involving nonlinear elements, the concept of limit cycles. A basic knowledge of sampled-data control systems including the z transform. The design of simple digital controllers. Application of the concepts of process dynamics and control to physiologic systems with particular attention to neural and homeostatic mechanisms.

Suggested Texts:

Coughanowr and Koppel, Process Systems Analysis and Control. 2nd Edition, McGraw Hill, 1991.

Luyben, W.L. Process Modelling, Simulation and Control for Chemical Engineers. 2nd Edition, McGraw Hill, N.Y. 1991.

04-Bio-A3 - Cellular and Molecular Biology and Biochemistry

Cell structure and function, including transport and chemical signals, adaptation of structure and function. Use of micro organisms in biotechnology. Biology of the prokaryotic cell. Chemical and physical structure of proteins, enzymes, nucleic acids, connective tissue and bone from molecular to microscopic levels. Relationship of chemical and physical structure of proteins to function including regulation of enzyme activity. Recombinant DNA technology including cloning, directed mutagenesis, DNA sequencing and expression of cloned genes. Development and use of recombinant proteins as therapeutic drugs. Fundamentals of therapeutic protein action. Site specific mutation of proteins. Protein-protein and protein-DNA interactions, receptor –ligand interactions, cell adhesion, cell migration, signal transduction, cell growth and differentiation. Post-

translational processing and secretion of proteins. Gene cloning and expression in mammalian cells. Techniques used for imaging, identification and measurement of biological materials.

Suggested Text:

Madigan, T., J. Martinko, and J. Parker, Brock Biology Of Microorganisms. Prentice-Hall, NJ. 2003.

04-Bio-A4 - Biomechanics

The musculoskeletal system; general characteristics and classification of tissues and joints. Elastic and viscoelastic mechanical characterization of biological tissues including bone, cartilage, ligament and tendon. Principles of viscoelastic and the rate sensitivity of biological materials. The stress-strain-time or constitutive equations for soft connective tissue components. Biomechanics and clinical problems in orthopaedics. Modelling and force analysis of musculoskeletal systems. Passive and active kinematics. Mechanical properties of biological and commonly used biomedical engineering materials.

Suggested Texts:

Berger, S.A., W. Goldsmith and E.R. Lewis, Introduction to Bioengineering. Oxford University Press, 2000.

Nordin, Margareta and Victor H. Frankel, Basic Biomechanics of the Musculoskeletal System. Lippincott Williams&Wilkins, 3rd edition, 2001.

04-Bio-A5 - Enzyme and Microbial Kinetics

Basic principles of bioprocessing fundamentals, which includes: kinetics of enzymatic reactions and microbial growth, batch and continuous cell growth kinetics, products formation and nutrient utilization, bioreactor systems. Basic principles of biochemical engineering. Applied enzyme catalysis, immobilized enzyme technology, kinetics of substrate utilization, product formation and biomass production in cell culture, batch and continuous culture. Applications of biochemical engineering.

Suggested Texts:

Blanch, H.W., D.S. Clark and Marcel Dekker, Biochemical Engineering. 1996.

Bailey, J.E. and E.F. Ollis, Biochemical Engineering Fundamentals. 2nd edition, McGraw Hill, 1986.

04-Bio-A6 - Anatomy and Physiology

Description of the human systems. Skeletal system with anatomy of superior members, inferior members and rachis. Osteoarticular system: physiology of bones, osseous tissues, articular cartilage, tendons, ligaments and muscles. Respiratory system, circulatory system, digestive system, urinary system, nervous system, reproductive apparatus. Structure-function relationships in human body systems.

Suggested Texts:

Guyton, Arthur C. and John E. Hall, Human Physiology and Mechanisms of Disease. 6th Ed., W.B. Saunders, Philadelphia, Pa., 1997.

Moffett, David F., Stacie B. Moffett, and Charles L. Schauf; Human Physiology, 2nd Ed.; Mosby, 1993.

04-Bio-A7 - Fluid Mechanics

Basics of momentum transfer and fluid flow; their application to the solution of engineering problems. Topics include: Engineering unit systems, dimensionless quantities; Basic concepts of fluid statics; Newton's law of viscosity; Steady and unsteady flow; Compressible and incompressible flow; Turbulent shear stress; Bernoulli's theorem, momentum transfer equations, equation of continuity; Computational fluid dynamics principles; Newtonian and Non –Newtonian fluids; External and internal flow; Fluid flow in pipes; Friction factors; Pumps, compressors, turbines; Flow measurement devices.

Suggested Text:

Middleman, S. An Introduction to Fluid Dynamics. Wiley, 1998.

04-Bio-A8 - Biophysical Measurements

Biomedical sensors and their application to the measurement of blood pressure, cardiac output and respiratory function. The origin of biopotentials including membrane and action potentials. Measurement of the electrocardiogram and the electroencephalogram. Basic electrode, biochemical sensor and laser applications including cardiac pacemakers and defibrillators. The basic concepts underlying computed transmission and emission tomography, magnetic resonance and ultrasound imaging. The imaging methods should be understood in terms of how imaging information is generated, detected and processed and how different hardware configurations and other factors affect image quality.

Suggested Text:

Webster, J.G. (Editor), Medical Instrumentation: Application and Design. 3rd Ed., Wiley, 1997.

04-Bio-A9 - Bioreactor Design

Transport phenomena in biochemical engineering systems, design and analysis of bioreactors, mixing, aeration, sterilization, instrumentation and control in bioprocesses. Internal and external mass transfer in immobilized systems. Oxygen mass transfer parameters of a bioreactor and design of an aeration system. Scale up of Bioprocesses.

Suggested Texts:

Blanch, H.W., D.S. Clark and Marcel Dekker, Biochemical Engineering. 1996.

Bailey, J.E. and E.F. Ollis, Biochemical Engineering Fundamentals. 2nd edition, McGraw Hill, 1986.

Aiba, S., A.E. Humphrey and N.F. Mills, Biochemical Engineering. 2nd edition, Academic Press, 1973.

Shuler, M.L. and F. Kargi, Biochemical Engineering Basic Concepts. Prentice Hall, 1992.

GROUP B - ELECTIVE EXAMINATIONS (THREE REQUIRED)

04-Bio-B1 - Biochemical Separations

The fundamentals of downstream separation and purification processes such as membrane separation processes, protein separation and purification and other separation processes of economic importance to the fermentation industry. Cell Disruption. Solid Liquid Separation, filtration, centrifugation. Membrane separation. Isoelectric focussing. Adsorption. Chromatography principles, Crystallization.

Suggested Texts:

Blanch, H.W., D.S. Clark and Marcel Dekker, Biochemical Engineering. 1996.

Shuler, M.L. and F. Kargi, Biochemical Engineering Basic Concepts. Prentice Hall, 1992.

04-Bio-B2 - Prostheses and Orthoses

Introduction, historic, terminology and classification of prostheses and orthoses. Partial or total replacement of limb or joint. Introduction to biomechanics related to design of prostheses and orthoses: clinical and mechanical aspects, biomaterials, biocompatibility. General design objectives and criteria. Design and assessment standards.

Suggested Text:

None at this time.

04-Bio-B3 - Biotransport Phenomena

Momentum, heat and mass transfer. Mass, linear momentum and energy balances. Differential analysis of laminar viscous flow. Differential analysis of heat conduction. Differential analysis of diffusion and convective transport. Biological examples of transport phenomena including: pharmacology and pharmacokinetics; absorption distribution, biotransformation, elimination, calculation of dosages; variability in drug response and adverse drug responses; drug delivery; microenvironment, transport and binding of small and large molecules; movement of cancer and immune cells; metastatic process, radiotherapy, chemotherapy, immunotherapy, hyperthermia, and photodynamic therapy of solid tumors. Numerical methods for computer simulation.

Suggested Texts:

Welty, James, Charles E. Wicks, Robert E. Wilson, and Gregory L. Rorrer, Fundamentals of momentum, Heat, and Mass Transfer. 4th Ed., Wiley, 2000.

Middleman, Stanley, An Introduction to Mass and Heat Transfer: Principles of Analysis and Design. Wiley, 1997.

04-Bio-B4 - Digital Image Processing

The extension of one dimensional sampling theory to two dimensions. Knowledge of the concepts of sampling geometry and sampling density. Two dimensional image transforms particularly the Fourier, Cosine and Walsh-Hadamard transforms. Important pixel operations for image enhancement particularly gray-scale modification and algebraic and geometric transforms. Convolution in two dimensions with particular application to image interpolation (upsampling). The spatial domain and frequency domain application of finite-extent point-spread filters for noise reduction, edge detection and image sharpening. Knowledge of the design and application of some common filters such as the Laplacian, the gradient and the Gaussian filters. Some knowledge of the concepts of image restoration from known degradations such as blur due to camera motion using some of the most common methods such as inverse and Wiener filtering and constrained deconvolution. The reconstruction of images from parallel and fan-beam projections as used in computed transmission tomography (CT).

Suggested Texts:

Gonzalez, R. and R. Woods, Digital Image Processing. 2nd Ed., Prentice Hall, 2002.

Suetens, P., Fundamentals of Medical Imaging. Cambridge University Press, 2002.

04-Bio-B5 - Cell and Tissue Engineering

Integration of relevant aspects of physiology, pathology, developmental biology, disease treatment and biomaterials to regenerative medicine in complex organ systems. Host response to tissue engineered constructs including complement, coagulation, immunological responses. Engineered replacements of kidney, lung, vascular, skin. Chemical, electrical, mechanical, materials, pathological and surgical aspects of construct development. Integrative exploration of the use of three-dimensional polymeric scaffolds and drug delivery vehicles, and gene therapy and cellular engineering for functional repair of injured tissues. Cell selection.

Suggested Text:

Lanza, R.P., R. Langer and W.L. Chick (eds), Principles of Tissue Engineering. 2nd edition, Academic Press, 2000.

04-Bio-B6 - Bioinstrumentation

Principles of design and analysis of electric instrumentation for biological applications. Ideal and non-ideal operational amplifiers, signal conditioning filters, sampling theory, analog to digital and digital to analog converters, sample and hold circuitry and multichannel data acquisition including the constraints imposed by real-time processing. The acquisition and processing of diagnostic signals such as the electrocardiogram, the echocardiogram, the blood pressure and hemoglobin oxygen saturation signals. Some basic knowledge of statistics for assessing the signal to noise characteristics of measured data.

Suggested Texts:

Webster, J.G. (Editor), Bioinstrumentation. Wiley, 2004

Webster, J.G. (Editor), Medical Instrumentation: Application and Design. 3rd Ed., Wiley, 1997.

04-Bio-B7 - Robotics and Manufacturing Automation

An overview of robotics and manufacturing technology and principles. Topics include: Automatic production and assembly, PLCs, sensors, actuators and drives, mechanization of part handling, industrial robots, and machine vision systems. Emphasis will be on the planning, design and implementation of automation systems.

Suggested Texts:

None at this time.

04-Bio-B8 - Rehabilitation Engineering

Introduction to rehabilitation engineering; Wheeled mobility: W/C history, technology and standards, fundamentals of manual W/Cs propulsion biomechanics, powered W/Cs and control systems; Functional disabilities: types of neuromuscular impairments; Specialized seating: classification of seating technologies, biomechanical principles of seating support & pressure, CAD/CAM seating applications; Hearing aids and cochlear implants: sensory and hearing aided technologies; Alternative & Augmentative Communication: rational, technologies & access strategies, principles of access & communication optimization; Prosthetics and orthotics: engineering principles of lower limb prostheses; ADL Devices: rational, design principles and use for upper & lower limb dysfunction; Measurement tools in rehabilitation engineering.

Suggested Texts:

Smith, Raymond V. & John H. Leslie, Rehabilitation Engineering. CRC Press, 1990.

Mann, William C. and Joseph P. Pane, Assistive Technology for Persons with Disabilities. The American Occupation Therapy Association Inc., 1991.

Webster, John G. et al, Electronic Devices for Rehabilitation. John Wiley & Sons, 1985.

04-Bio-B9 - Artificial Intelligence and Expert Systems

AI-based decision making in biology and medicine using predicate calculus, structures and strategies for state space search, heuristic search and stochastic methods. Knowledge representation, reasoning and decision-making under uncertainty as well as case-based reasoning, decision trees. Rule-based and expert systems, inference mechanisms and knowledge engineering. Machine learning including supervised learning, self-organization, reinforcement learning and evolutionary computing. Intelligent biomedical information systems, intelligent devices and instruments such as interactive implants and replacements and measurement systems. Automated reasoning and data mining. Advanced methods for problem solving including natural language processing, planning and perception.

Suggested Texts:

Russell, S. and P. Norvig, Artificial Intelligence: A Modern Approach. 2nd Edition. Prentice Hall, 2003. ISBN: 0137903952

Luger, G., Artificial Intelligence: Structures and Strategies for Complex Problem Solving. 5th Ed., Addison Wesley, 2005.

04-Bio-B10 - Analytical Biochemistry

Relevant analytical techniques for characterization of biological systems and materials. Nuclear magnetic resonance. Fourier transform infra red analysis. SDS-PAGE and Western blotting. HPLC. Flow cytometry. DNA gel extraction and ligation. Plasmid DNA mini-preps and PCR. Affinity purification and electrophoresis. Surface analysis techniques including x-ray photoelectron spectroscopy, atomic force microscopy, interfacial tension and ellipsometry.

Suggested Texts:

Mikkelsen, Susan R. and Eduardo Corton, Bioanalytical chemistry. Wiley Interscience, 2004. ISBN: 0-471-54447-7

Holme, D.J. and H. Peck, Analytical Biochemistry. 3rd ed., Longman, 1998.

04-Bio-B11 - Ergonomics

Basic human abilities and characteristics, including vision and hearing. Psychomotor characteristics. Anthropometry: static and dynamic human body dimensions and muscle strength. Environmental factors, including illumination, atmospheric conditions, noise, and vibration. Ergonomic work design, including layout of equipment, manual work aids, design of seating, and person-machine interfaces: instruments, controls, and software.

Suggested Texts:

Bridger, R.S., Introduction to Ergonomics. McGraw-Hill, 1995. ISBN 0-07-007741-X.

Kodak Ergonomics Group, Ergonomic Design for People at Work, Volumes I and II. Van Nostrand Reinhold Co. Ltd., 1986.

04-Bio-B12 - Applied Optics/Photonics

Basic optics of rays; reflection, refraction, and polarization. Lens systems and image formation. Principles of basic optical instruments such as magnifiers, microscopes and telescopes. Basics of light sources: lasers, light emitting diodes, thermal light sources, fluorescence, and photodetectors. Tissue optics and light-tissue interactions and dosimetry. Principles of fibre optics and light guides, endoscopic systems and applications. Biomedical applications of photonics such as phototherapy and photodiagnosis, tissue oximetry, optical spectroscopy and microscopy, fluorescence marking.

Suggested Text:

Prasad, N., Introduction to Biophotonics. Wiley, 2004.